

Arpi Abrahamian

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OBJECTIVE

Aspiring mechanical engineer with hands-on experience in fabrication, electronics, and interdisciplinary design. Looking to contribute to collaborative engineering projects and apply technical skills to real-world challenges.

EDUCATION

Cornell University, Duffield College of Engineering

Ithaca, NY

B.S. Mechanical Engineering

Expected May 2029

Relevant Coursework: Statics, Physics Mechanics, E&M, Differential Equations, Multivariable Calculus, Python

EXPERIENCE

Bovay Laboratory Complex, Cornell University

Ithaca, NY

Student Research/Field/Lab Assistant

January 2026 – Present

- Complete assigned independent technical projects with minimal supervision.
- Design and fabricate custom test fixtures and experimental components as needed.
- Support civil & environmental research through specimen prep, instrumentation calibration, & lab operations.
- Collaborate with faculty, staff, and lab personnel to support coursework, workshops, and research projects.

International Outreach Program, HoPE (Hands-on Physics Experience)

Pescara–Maglie–Ostia, Italy

Mentor Facilitator

January 2026

- Led groups of 100+ students in constructing 15-ft pyramidal structures.
- Taught classes on heat transfer and electromagnetic waves (ages 5–19).
- Guided mechanical and electrical design projects through ideation, design, and creation.

Engineering Design Workshop, Massachusetts Institute of Technology

Cambridge, MA

Project Mentor

July 2023 – August 2025

- Designed and deployed a GPS/temperature sensor array for Charles River heat mapping.
- Led Arduino-based design projects (automated folding machine, musical water fountain)
- Fabricated 20+ cast copper rings and sterling silver pins
- Taught electronics and circuits through household item disassembly, led soldering workshops
- Assisted in assembly and live operation of a hydraulophone and fire organ

Molecular Biology Curriculum Kit Assembly, MIT

Cambridge, MA

Lab Kit Assembly Assistant

July 2025

- Assembled 60+ molecular biology curriculum kits with DNA, RNA, and tRNA lessons.
- Sorted, labeled, and organized all components to specification for classroom use.
- Recognized for reliability and attention to detail; trusted to work independently and deliver consistent results.

FIRST Robotics, Team 5735

Wayland, MA

Electrical Lead

Oct 2023 – May 2025

- Designed 2-stage telescoping climber with custom brackets and Hall-effect sensing.
- Iterated 7+ CNC redesigns; operated and repaired system under competition conditions.
- Trained and mentored 5+ junior team members in wiring, soldering, and system testing.

SKILLS AND INTERESTS

Computing & Digital Tools: Java, Python, Arduino, HTML; CAD (SolidWorks, Onshape, Fusion 360)

Fabrication: Laser cutting, soldering, 3D printing, CNC routing, casting, mold design

Tools & Machinery: Drill press, chop saw, jigsaw, belt sander, blade runner saw, embroidery machine, heat shrink